Krzysztof Mazurek

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Summary

Enthusiastic about modeling, simulation, and control of dynamic mechanical systems, with interests in topics such as: GNC, mission analysis, mechanical systems dynamics. I aim to deepen my expertise in numerical computing, software development and optimization.

Education

Politecnico di Milano Sep. 2025 - Present (expected - 2027)

Master's degree in Space Engineering

- Operations Research, Orbital Mechanics, Spacecraft Attitude Dynamics, Space Systems Engineering and Operations, Electromagnetics and Signal Processing for Spaceborne Applications, Space Propulsion

Warsaw University of Technology Oct. 2021 - Mar. 2025

Robotics and Automatic Control - Final grade : 4.4 / 5

Professional experience

Structural Analysis Engineer May 2025 - Aug. 2025

Sener Aerospace & Defence *Warsaw, Poland*

- Performed static and modal FEA of structures (ESA COMET container, ARIEL satellite components) using Siemens Femap.
- Automated FEA workflows using VBA and Python: temperature mesh mapping from Esatan to Femap, result postprocessing, midsurface preprocessing.

Software Engineer Jul. 2024 - Aug. 2024

Sellita Watch Company *Le-Chaux de Fonds, Switzerland*

- Developed software module for Beckhoff PLC in TwinCat3- movement control of ultra-precise servo press for assembling watch components based on the user-specified criteria.
- Developed HMI for the machine control module in C#.Net framework.

Student activities

Electrical team coordinator, Mechanical team member Jan. 2022 - Jun. 2025

Student Association for Vehicle Aerodynamics - MotoStundent Electric 2025 *Warsaw, Poland*

- Responsible for early-stage development of the electric drivetrain- design and validation.
- Simulated and improved the shape of a motorcycle rear pro-link suspension in Msc Adams.
- Designed a steel-tubed swingarm, chain tensioner assembly, rear brake adapter using Siemens NX.

3D printing Science Club vice-president Oct. 2022 - Oct. 2024

Warsaw University of Technology *Warsaw, Poland*

- Designed and manufactured models, customized and assembled machines (CNC, resin/ FDM printers).

University & personal projects

Reverse engineering of the JAXA GOSAT mission Mar. 2026 - Jun. 2026

2D static and dynamical FEM solver Mar. 2026 - Mar. 2026

Spacecraft attitude dynamics - modeling and control Sep. 2025 - Dec. 2025

Orbital mechanics - Interplantery mission and perturbation analysis Sep. 2025 - Dec. 2025

Feedforward control of underactuated multibody systems in a co-simulation Oct. 2024 - Mar. 2025

Satellite solar array opening simulation Mar. 2024 - Jun. 2024

Technical Skills & Foreign language

Programming: Matlab & Simulink, Python, , C++
Engineering Software: Msc Adams, Siemens Femap, Ansys Mechanical & Spaceclaim, Siemens NX, Fusion 360
Tools: Git, GitHub, Linux, LaTeX, MS Office, TwinCat3
Foreign language: English C1+, German B1, Polish native